Second Edition

Digital Control of Dynamic Systems

Gene F. Franklin
Stanford University

J. David Powell
Stanford University

Michael L. Workman
IBM Corporation

ADDISON-WESLEY PUBLISHING COMPANY
Reading, Massachusetts • Menlo Park, California • New York
Don Mills, Ontario • Wokingham, England • Amsterdam
Bonn • Sydney • Singapore • Tokyo • Madrid • San Juan

Contents

CHAPTER 1
Introduction 1

1.1 Problem Definition 1
1.2 Example Systems for Study 5
1.3 Overview of Design Approach 7
1.4 Computer-Aided Design 9
1.5 Summary 10
1.6 Suggestions for Further Reading 10
Problems and Exercises 11

CHAPTER 2
Linear, Discrete, Dynamic-Systems Analysis: The z-Transform 13

2.1 Introduction 13
2.2 Linear Difference Equations 13
2.3 The Discrete Transfer Function 19
2.3.1 The z-Transform 19
2.3.2 The Transfer Function 20
2.3.3 Block Diagrams and State-Variable Descriptions 23
2.3.4 Relation of Transfer Function to Pulse Response 30
2.3.5 External Stability and Jury's Test 35
2.4 Discrete Models of Sampled-Data Systems 41
2.4.1 Using the z-Transform 41
2.4.2 Continuous Time Delay 44
2.4.3 State-Space Form 46
2.4.4 State-Space Models for Systems with Delay 56
2.4.5 Numerical Considerations in Computing Φ and Γ 61

2.5 Signal Analysis and Dynamic Response 64
2.5.1 The Unit Pulse 65
2.5.2 The Unit Step 65
2.5.3 Exponential 66
2.5.4 General Sinusoid 67
2.5.5 Correspondence with Continuous Signals 71
2.5.6 Step Response 73
2.6 Frequency Response 77
2.7 Properties of the z-Transform 82
2.7.1 z-Transform Properties 82
2.7.2 Another Derivation of the Transfer Function 92
2.8 Summary 95
Problems and Exercises 95

CHAPTER 3
Sampled-Data Systems 101
3.1 Introduction 101
3.2 Analysis of the Sample and Hold 102
3.3 Spectrum of a Sampled Signal and Aliasing 107
3.4 Data Extrapolation and Impostors 112
3.5 Block-Diagram Analysis 121
3.6 Summary 130
Problems and Exercises 130

CHAPTER 4
Discrete Equivalents to Continuous Transfer Functions: The Digital Filter 133
4.1 Introduction 133
4.2 Design of Discrete Equivalents by Numerical Integration 135
4.3 Zero-Pole Mapping Equivalents 147
4.4 Hold Equivalents 149
4.4.1 Zero-Order Hold Equivalent 149
4.4.2 Triangle Hold Equivalent 151
4.5 Summary 155
Problems and Exercises 156

CHAPTER 5
Design of Digital Control Systems Using Transform Techniques 158
5.1 Introduction 158
5.2 Control System Specifications 159
5.3 Design Using Emulation 174

5.4 z-Plane Design Using Root Locus 183
5.5 Frequency-Response Methods with the z-Transform 193
5.5.1 Gain and Phase Plotting 194
5.5.2 Nyquist Stability 197
5.5.3 Low-Frequency Gains and Error Coefficients 203
5.5.4 Compensator Design 205
5.6 Frequency-Response Methods with the w-Transform 210
5.7 Direct Design Method of Ragazzini 216
5.8 PID Control 222
5.8.1 Proportional Control 223
5.8.2 Derivative Control 223
5.8.3 Integral Control 224
5.8.4 PID Control 224
5.8.5 Ziegler-Nichols PID Tuning 224
5.9 Summary 229
Problems and Exercises 230

CHAPTER 6
Design of Digital Control Systems Using State-Space Methods 238
6.1 Introduction 238
6.2 Control-Law Design 239
6.2.1 Pole Placement 241
6.2.2 Controllability 244
6.2.3 Ackermann's Formula 245
6.3 Estimator Design 250
6.3.1 Prediction Estimators 251
6.3.2 Observability 255
6.3.3 Ackermann's Formula 255
6.3.4 Current Estimators 256
6.3.5 Reduced-Order Estimators 261
6.4 Regulator Design: Combined Control Law and Estimator 264
6.4.1 The Separation Principle 264
6.4.2 Guidelines for Pole Placement 271
6.5 Introduction of the Reference Input 273
6.5.1 Reference Inputs for Full-State Feedback 273
6.5.2 Reference Inputs with Estimators: The State-Command Structure 278
6.5.3 Effect of Reference Input on System Zeros 280
6.5.4 Output Error Command 284
6.5.5 A Comparison of the Estimator Structure and Classical Methods 287
6.6 Integral Control 289
6.6.1 Integral Control by State Augmentation 289
6.6.2 Bias Estimation 292

6.7 Pole Placement using Polynomials 295
6.7.1 Integral Control 297
6.8 Controllability and Observability 301
6.9 Summary 308
Appendix to Chapter 6 309
Problems and Exercises 315

CHAPTER 7
Quantization Effects 322
7.1 Introduction 322
7.2 Analysis of Round-Off Error 322
7.3 Effects of Roundoff of Parameters 336
7.4 Limit Cycles and Dither 340
7.5 Summary 345
Problems and Exercises 345

CHAPTER 8
System Identification 349
8.1 Introduction and Problem Definition 349
8.2 Identification of Nonparametric Models 353
8.3 Models for Parametric Identification 366
8.4 Least Squares 374
8.5 Recursive Least Squares 378
8.6 Stochastic Least Squares 383
8.7 Maximum Likelihood 395
8.8 Numerical Search for the Maximum-Likelihood Estimate 402
8.9 Summary 413
Problems and Exercises 413

CHAPTER 9
Multivariable and Optimal Control 417
9.1 Introduction 417
9.2 Decoupling 418
9.3 Time-Varying Optimal Control 422
9.4 LQR Steady-State Optimal Control 430
9.4.1 Reciprocal Root Properties 432
9.4.2 Symmetric Root Locus 433
9.4.3 Eigenvector Decomposition 435
9.4.4 Cost Equivalents 439
9.4.5 Emulation by Equivalent Cost 441

9.5 Optimal Estimation 444
9.5.1 The Kalman Filter 444
9.5.2 Steady-State Optimal Estimation 451
9.5.3 Noise Matrices and Discrete Equivalents 454
9.6 Multivariable Control Design 458
9.6.1 Selection of Weighting Matrices Q_1 and Q_2 458
9.6.2 Pincer Procedure 459
9.6.3 Paper-Machine Design Example 461
9.6.4 Magnetic-Tape-Drive Design Example 465
9.7 Summary 480
Problems and Exercises 480

CHAPTER 10
Sample Rate Selection 483
10.1 Introduction 483
10.2 The Sampling Theorem's Limit 484
10.3 Time Response and Smoothness 485
10.4 Errors Due to Random Plant Disturbances 488
10.5 Sensitivity to Parameter Variations 497
10.6 Measurement Noise and Antialiasing Filters 501
10.7 Multirate Sampling 506
10.8 Summary 511
Problems and Exercises 513

CHAPTER 11
Nonlinear Control 516
11.1 Introduction 516
11.2 Analysis Techniques 516
11.2.1 Simulation 517
11.2.2 Linearization 523
11.2.3 Describing Functions 535
11.2.4 Equivalent Gains 551
11.2.5 Circle Criterion 555
11.2.6 Lyapunov Functions 558
11.3 Nonlinear Control Structures: Design 561
11.3.1 Large-Signal Linearization: Inverse Nonlinearities 562
11.3.2 Time-Optimal Servomechanisms 581
11.3.3 Extended PTOS for Flexible Structures 596
11.3.4 Introduction to Adaptive Control 602
11.3.5 Controller versus Regulator 615
11.3.6 Adaptive Nonlinear Control 618
11.4 Design with Nonlinear Cost Functions 625
11.4.1 Random Neighborhood Search 625

11.5 Summary 633
Problems and Exercises 635

CHAPTER 12
Application and Practice of Digital Control 641
12.1 Introduction 641
12.2 Digital Control System Hardware 642
12.2.1 Processors 643
12.2.2 Support Hardware 650
12.2.3 Error Detection and Correction Hardware 650
12.2.4 D/A's, A/D's, and Sample and Hold's 653
12.2.5 Saturating Components 662
12.2.6 Sensors 664
12.3 Software or Firmware Development 666
12.3.1 More than Control Equations 667
12.3.2 Microcode Versions 668
12.3.3 Reprogramming versus Changing Hardware 668
12.3.4 Processor Microcode Development Tools 669
12.3.5 Error Detection and Correction Microcode 671
12.3.6 Control Wind-Up 672
12.3.7 Fixed- and Floating-Point Arithmetic 676
12.3.8 Fixed-Point Arithmetic Scaling 679
12.3.9 Floating-Point Arithmetic Scaling 680
12.4 System Integration 681
12.4.1 Hardware Simulators 681
12.4.2 Frequency Response and Modal Analysis 683
12.5 Manufacturing Digital Control Systems 691
12.5.1 Communication Interface 691
12.5.2 Robust Design 693
12.5.3 Design Specifications 694
12.5.4 Self-Test 695
12.5.5 Adjustments 697
12.6 Program Structure 698
12.6.1 Minimizing Control Delay 700
12.6.2 Sample Rate Selection Revisited 701
12.7 Case Design: Disk-Drive Servo 703
12.7.1 Disk Drives 703
12.7.2 Goals and Objectives 724
12.7.3 Design 728
12.7.4 Implementation 742
12.8 Summary 746
12.9 Problems and Exercises 747
Appendix A to Chapter 12 751
Appendix B to Chapter 12 755

APPENDIX A Examples 777

A.1 Single-Axis Satellite Attitude Control 777
A.2 A Servomechanism for Antenna Azimuth Control 779
A.3 Temperature Control of Fluid in a Tank 783
A.4 Control through a Flexible Structure 785
A.5 Control of a Pressurized Flow Box 788

APPENDIX B Tables 790

B.1 Properties of z-Transforms 791
B.2 Table of z-Transforms 792
B.3 w-Plane Transfer Functions 794

APPENDIX C A Few Results from Matrix Analysis 796

C.1 Determinants and the Matrix Inverse 796
C.2 Eigenvalues and Eigenvectors 798
C.3 Similarity Transformations 800
C.4 The Caley-Hamilton Theorem 803

APPENDIX D Summary of Facts from the Theory of Probability and Stochastic Processes 805

D.1 Random Variables 805
D.2 Expectation 807
D.3 More than One Random Variable 810
D.4 Stochastic Processes 812

APPENDIX E CAD Routines 817

References 819

Index 833